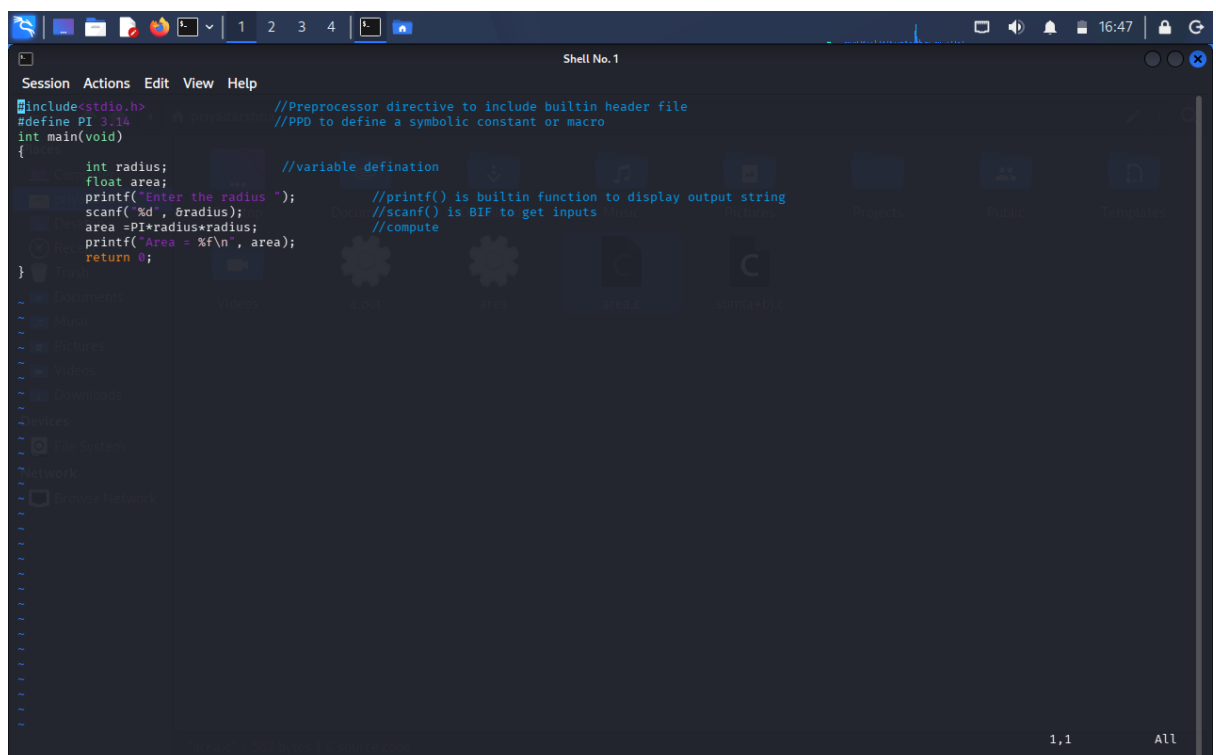


EXPERIMENT -- 2

Title : C Programming Environment

1. Write a C program to find the area of circle . Attach a snippet.



The screenshot shows a code editor window titled "Shell No. 1" with a menu bar (Session, Actions, Edit, View, Help) and a toolbar. The code is as follows:

```
#include<stdio.h> //Preprocessor directive to include builtin header file
#define PI 3.14 //PPD to define a symbolic constant or macro
int main(void)
{
    int radius; //variable definition
    float area;
    printf("Enter the radius "); //printf() is builtin function to display output string
    scanf("%d", &radius); //scanf() is BIF to get inputs
    area =PI*radius*radius; //compute
    printf("Area = %f\n", area);
    return 0;
}
```

The left sidebar shows a file explorer with folders like Documents, Downloads, and Desktop. The bottom right corner shows the cursor position "1,1" and the word "All".

2. Demonstrate stages of compilations of the above program with -save-temps option used with cc/gcc command. Write a brief notes on what happens in each stages with suitable snippet.

Stages of complitios-

1 Preprocessing—Converts circle.c into circle.i . Header files are included , macros are expanded , and it commnets are removed.

2 Compilation- It converts circle.i into the assembly file circle.s.

3 Assembly – It converts circle.s into the object file circle.o ,
containing machine code.

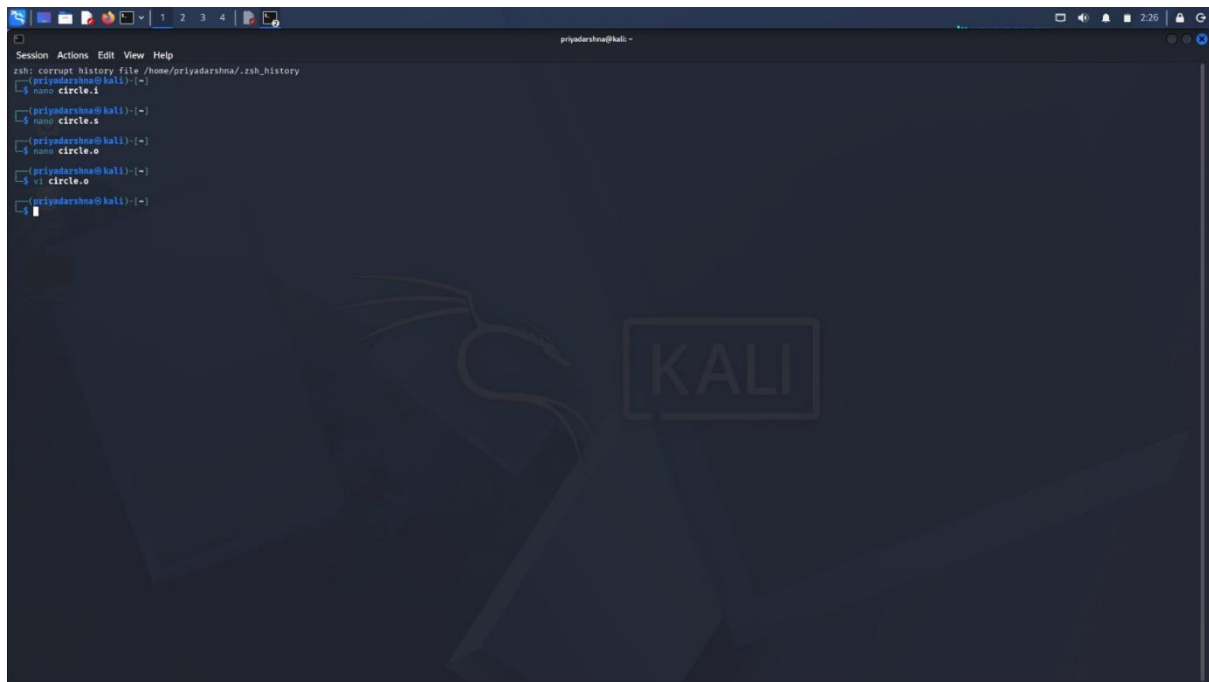
4 Linking – Links circle.o with the required libraries and creates the
executable file circle.

```
Session Actions Edit View Help
zsh: corrupt history file /home/priyadarshna/.zsh_history
(priyadarshna@kali)~$ nano circle.c
(priyadarshna@kali)~$ gcc -save-temps circle.c -o circle
(priyadarshna@kali)~$ ls -l circle*
-rwxrwxr-x 1 priyadarshna priyadarshna 16016 Aug 20 01:42 circle
-rw-rw-r-- 1 priyadarshna priyadarshna 652 Aug 20 01:41 circle.c
-rw-rw-r-- 1 priyadarshna priyadarshna 0 Aug 20 01:35 circle.c.save
-rw-rw-r-- 1 priyadarshna priyadarshna 630 Aug 20 01:36 circle.c.save.1
-rw-rw-r-- 1 priyadarshna priyadarshna 21595 Aug 20 01:42 circle.i
-rw-rw-r-- 1 priyadarshna priyadarshna 1688 Aug 20 01:42 circle.o
-rw-rw-r-- 1 priyadarshna priyadarshna 1131 Aug 20 01:42 circle.s
(priyadarshna@kali)~$ ./circle
Enter the radius : 6
Area = 113.040001
(priyadarshna@kali)~$
(priyadarshna@kali)~$
(priyadarshna@kali)~$ file circle.c circle.i circle.s circle.o circle
circle.c: C source, ASCII text
circle.i: C source, ASCII text
circle.s: assembler source, ASCII text
circle.o: ELF 64-bit LSB relocatable, x86-64, version 1 (SYSV), not stripped
circle: ELF 64-bit LSB pie executable, x86-64, version 1 (SYSV), dynamically linked, interpreter /lib64/ld-linux-x86-64.so.2, BuildID[sha1]=163ccce7fa29d87a23bed6a789d95a5face7369, for GNU/Linux 3.2.0, not stripped
(priyadarshna@kali)~$
(priyadarshna@kali)~$ xfce4-screenshooter -w
```

```
Session Actions Edit View Help
zsh: corrupt history file /home/priyadarshna/.zsh_history
~$ nano circle.c
~$ gcc -save-temps circle.c -o circle
~$ ls
a.out area.c circle.c circle.c.save circle.o Desktop Downloads hello.c linux_lab output.txt Pictures Public test.txt
area circle circle.c.save circle.i circle.s Documents Hello linux_assignment Music output.txt Projects Templates Videos
~$ xfreer-screenshooter -w
```

3. Verifying human readability of the generated intermediate codes *.i, *.s, and *.o files while opening them in linux editor (vi/gedit). Attach suitable snippets.

- .i file: Human readable : it contains preprocessed C code with included headers and explain macros.
- .s file: Human readable: it contains assembly language instructions but requires assembly knowledge.
- .o file: Not human readable: it contains binary machine code.

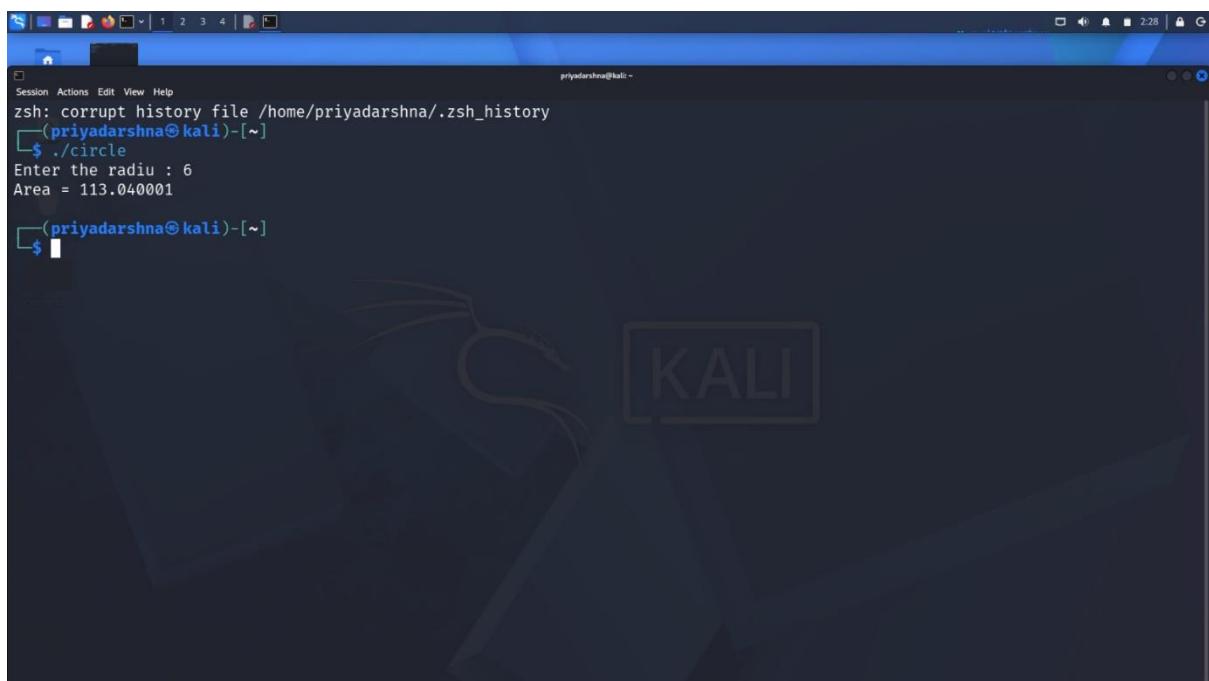


A terminal window on a Kali Linux system. The window title is 'priyadarshna@kali -'. The terminal shows the following commands and output:

```
zsh: corrupt history file /home/priyadarshna/.zsh_history
(priyadarshna@kali)~$ nano circle.i
(priyadarshna@kali)~$ nano circle.s
(priyadarshna@kali)~$ nano circle.o
(priyadarshna@kali)~$ vi circle.o
(priyadarshna@kali)~$
```

The background of the terminal features a faint Kali Linux logo and the word 'KALI' in a box.

4 Execution of a.out file in linux command line (using ./.) . Attach snippet.



A terminal window on a Kali Linux system. The window title is 'priyadarshna@kali -'. The terminal shows the following commands and output:

```
zsh: corrupt history file /home/priyadarshna/.zsh_history
(priyadarshna@kali)~$ ./circle
Enter the radiu : 6
Area = 113.040001
(priyadarshna@kali)~$
```

The background of the terminal features a faint Kali Linux logo and the word 'KALI' in a box.

NAME – PRIYADARSHNA

BATCH – 15

SAPID – 590032928

TOPIC – C PROGRAMMING ENVIRONMENT